STAFF REVIEW · H446 223 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It never overestimates the true cheapest remaining cost.

2. Places it at its final index with smaller/equal values to its left.

3. The first node in a linked list.

4. Otherwise the original successor can become unreachable and part of the list may be lost.

I Do — source worked steps

1. Predict first — Set newNode.Next to the existing successor 9.

2. Model the reasoning — Update node4.Next to reference the new 7 node. The 9→12 remainder is preserved.

We Do — guided CHECK

1. Model the reasoning
Then change node2.Next to reference 6; the final chain is 2→6→8→11.

You Do — source mark schemes

1. [3] Head is the node containing 3 [1]; 5.Next.Value is 8 [1]; 8.Next is null [1].

2. [5] Node inserted = new(12,node8.Next); [2]; node8.Next = inserted; [1]; 8→12→20→30 [2].

3. [6] Set node10.Next=node10.Next!.Next [2] to bypass 20 [1], leaving 10→30 [1]; finding node10 from head may take O(n) [1], but changing the link once identified is O(1) [1].

Exit Ticket — source answer

1. False. Insertion changes references; later node values do not need shifting.